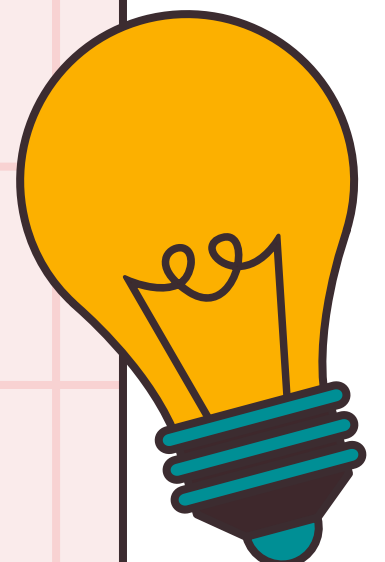
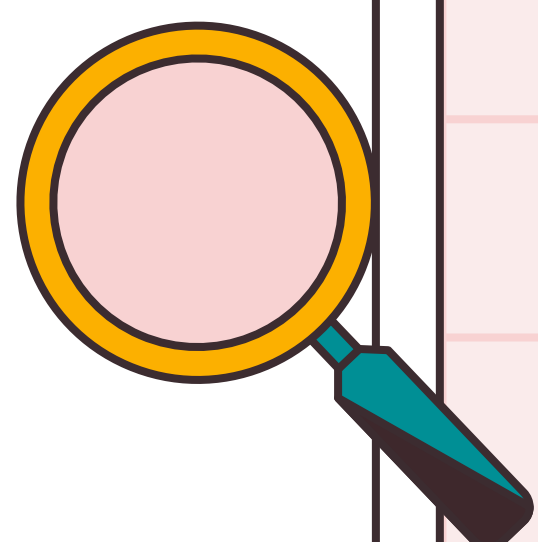




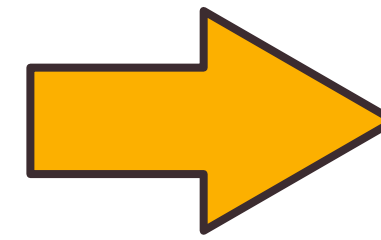
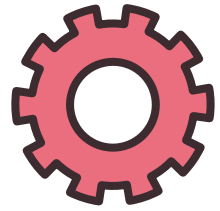
**Final Project**

# Medical Insurance Cost Prediction



From raw data to deployable model

ABBA ABDELDJABBAR - 10 MAY 2026 - Data Analysis



01

# Objective & Dataset



# Business Problem & Data



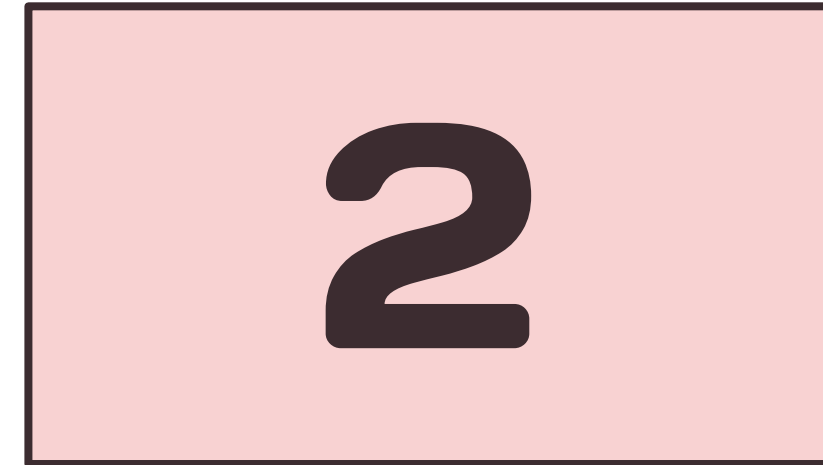
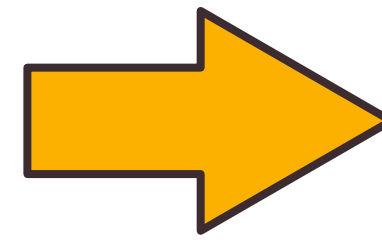
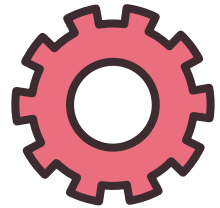
**Goal:** Predict annual medical charges to help insurance companies set premiums.

**Dataset:** insurance.csv

- 1,338 patients, 7 features
- **Target:** charges (USD)

**numeric :** age, bmi, children

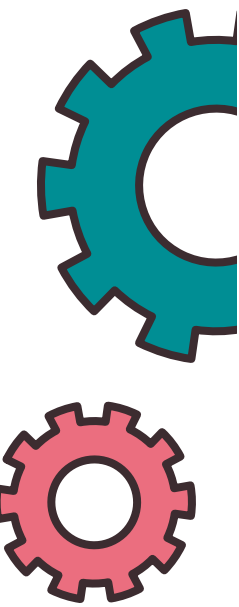
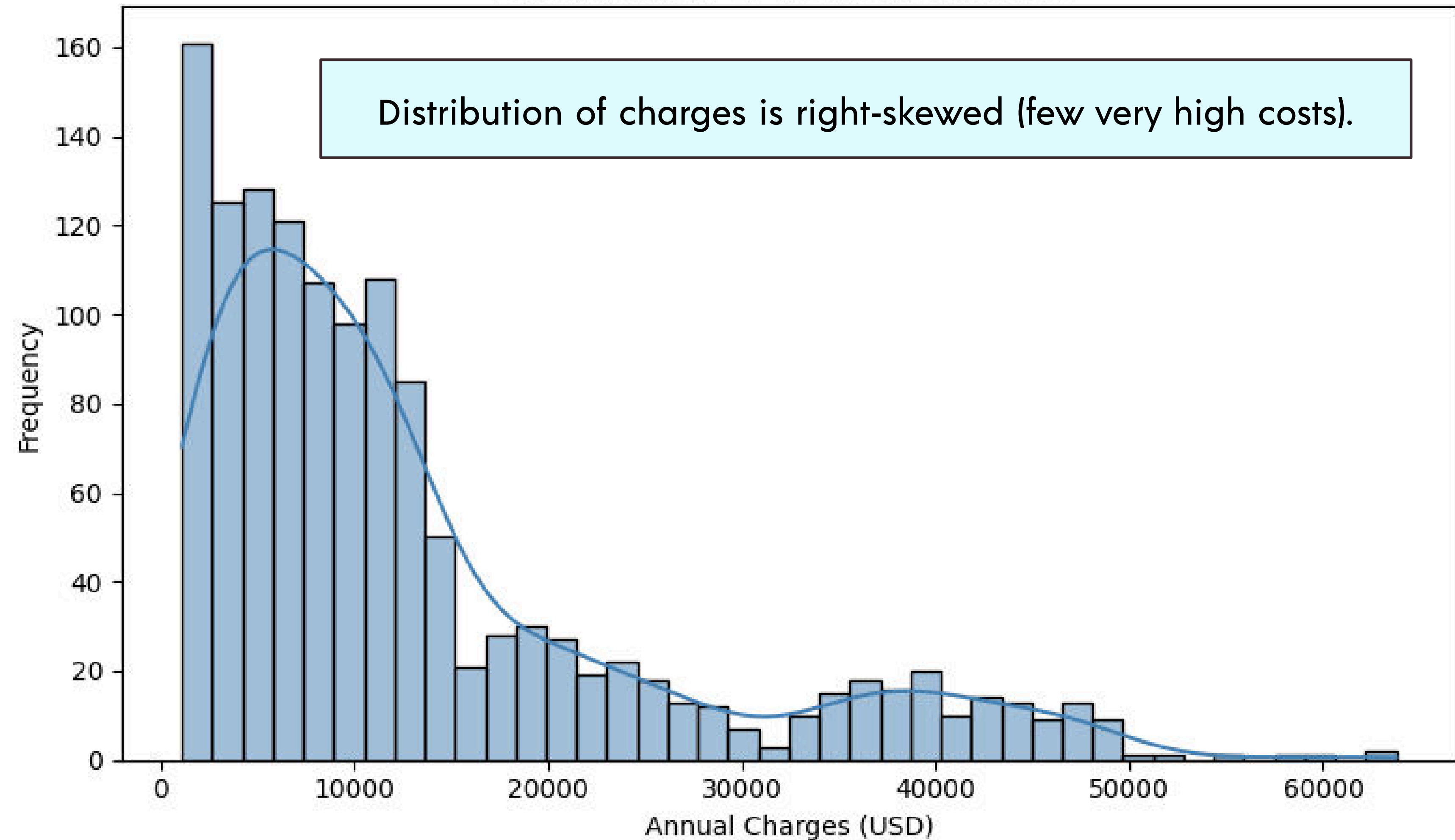
**categorical :** sex, smoker, region



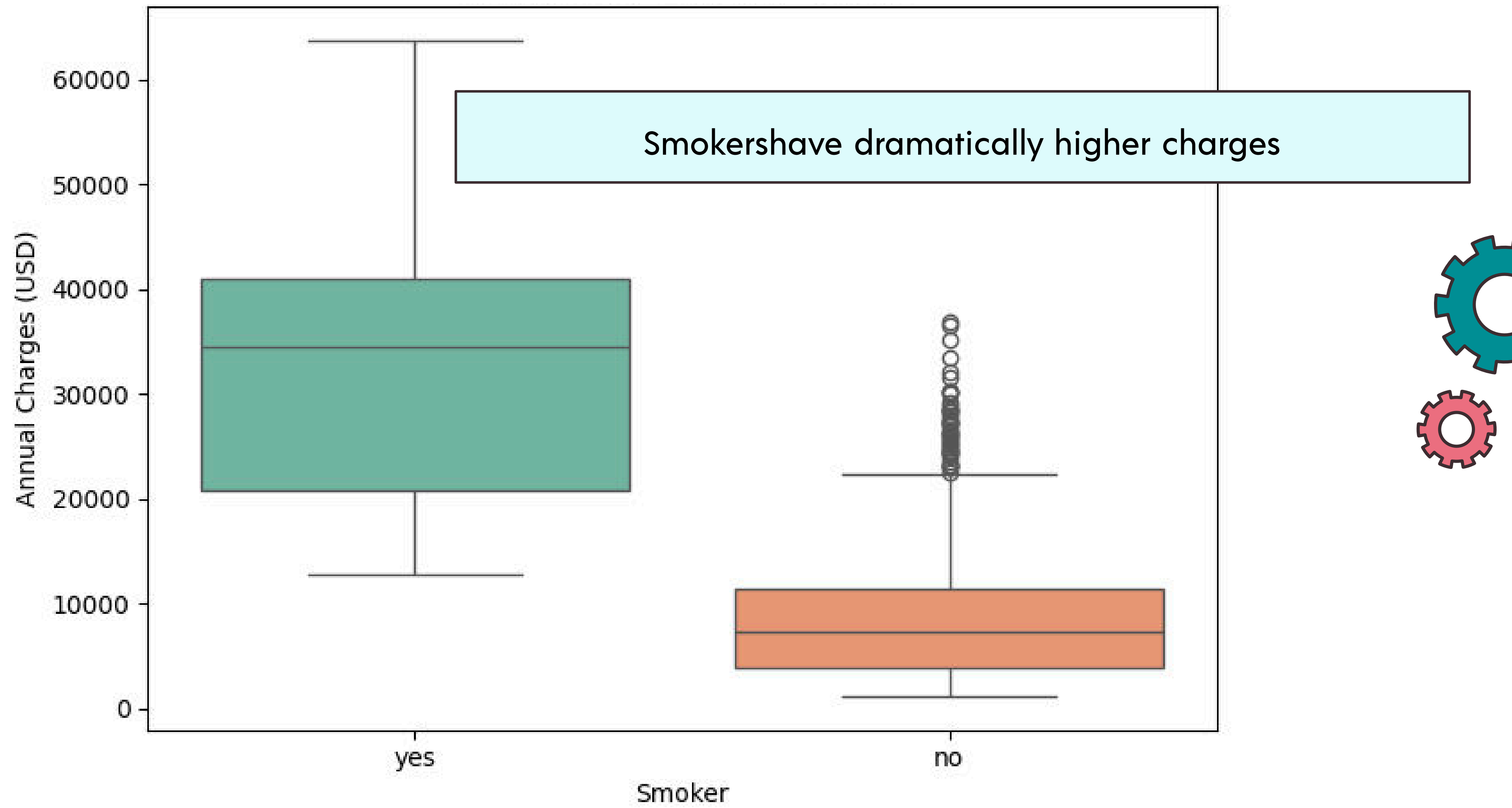
# EDA & Key Findings



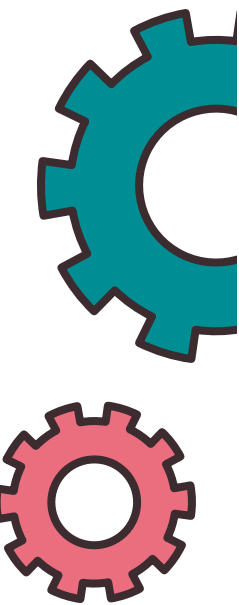
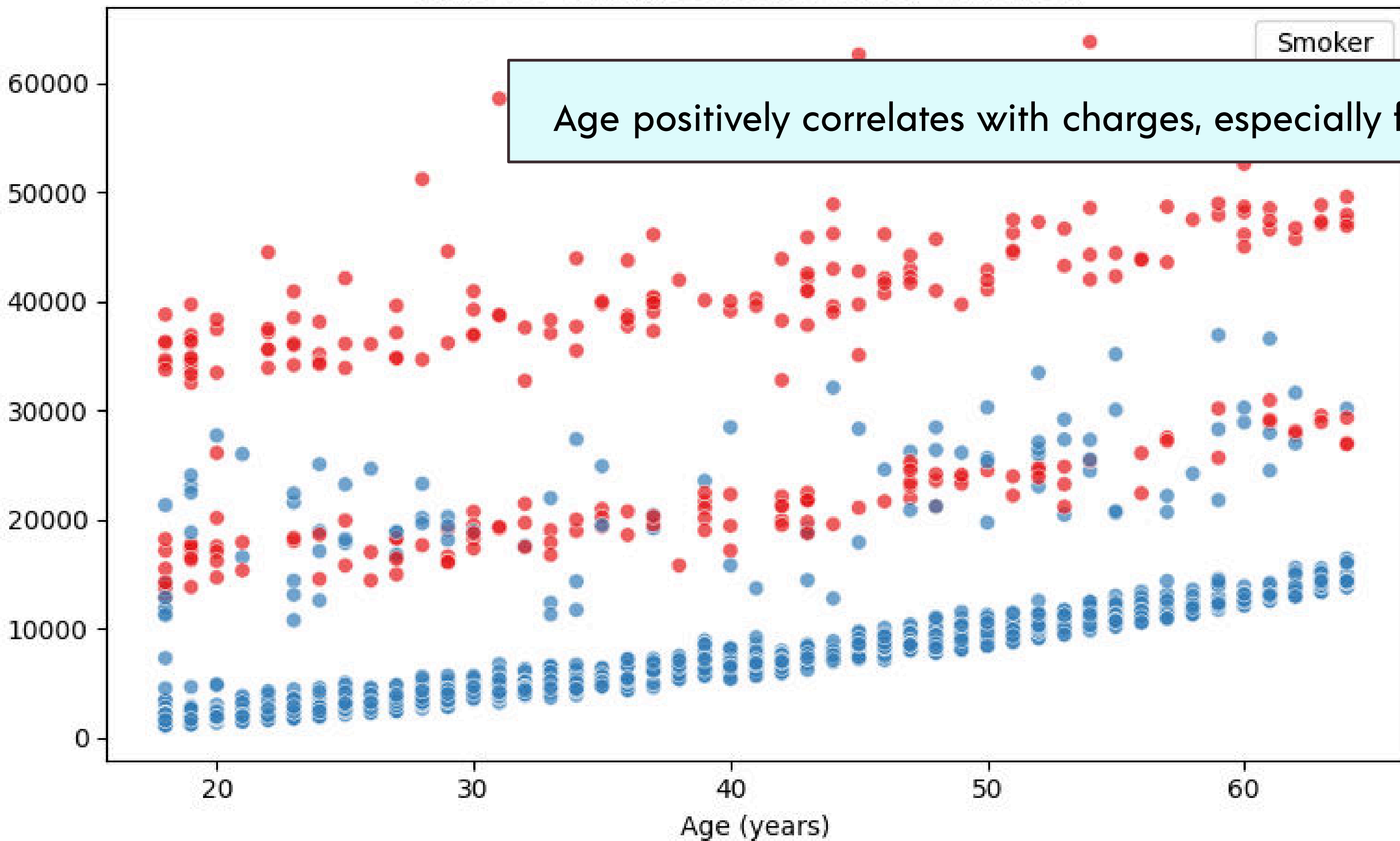
# Distribution of Medical Charges



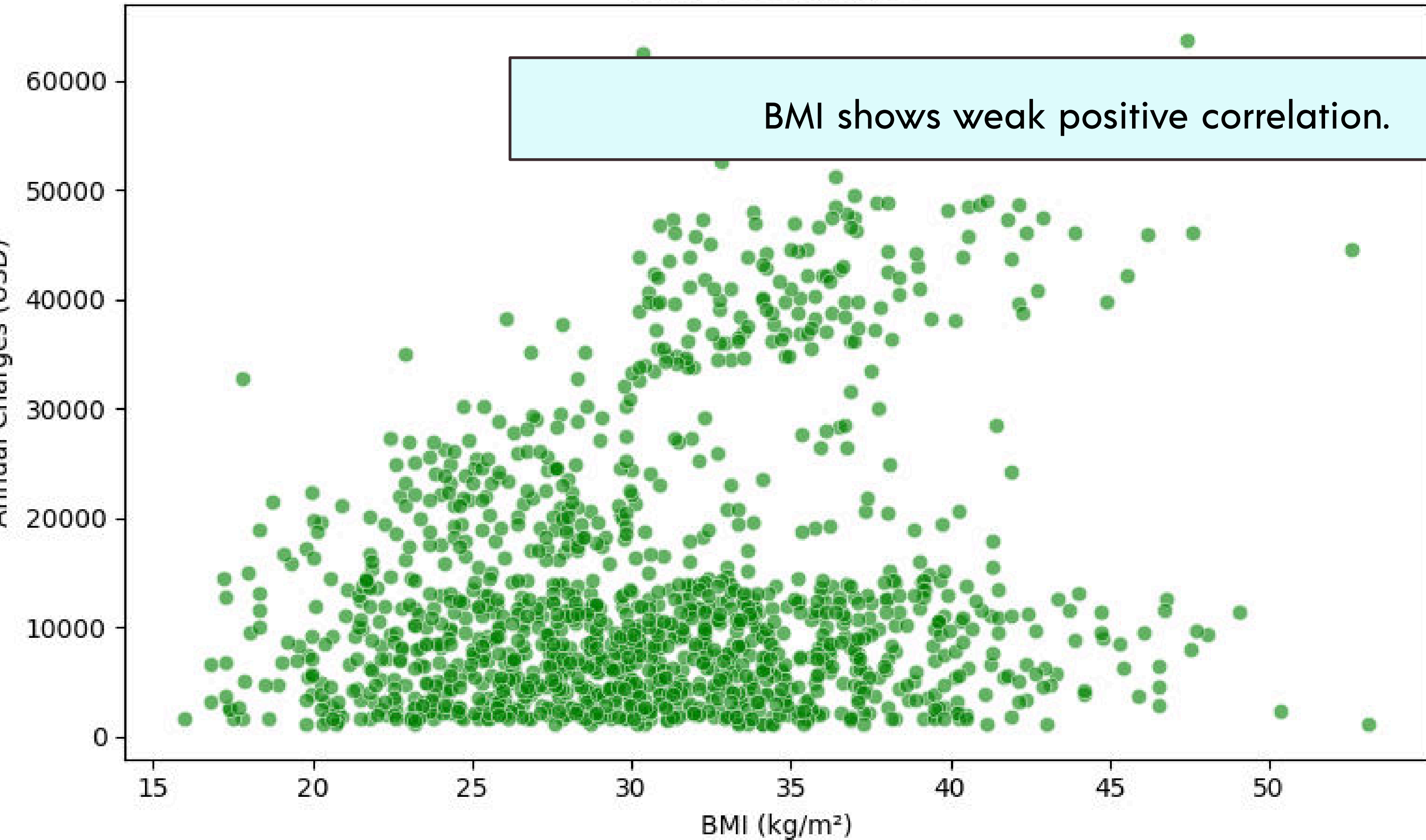




Age vs Charges (Colored by Smoker)

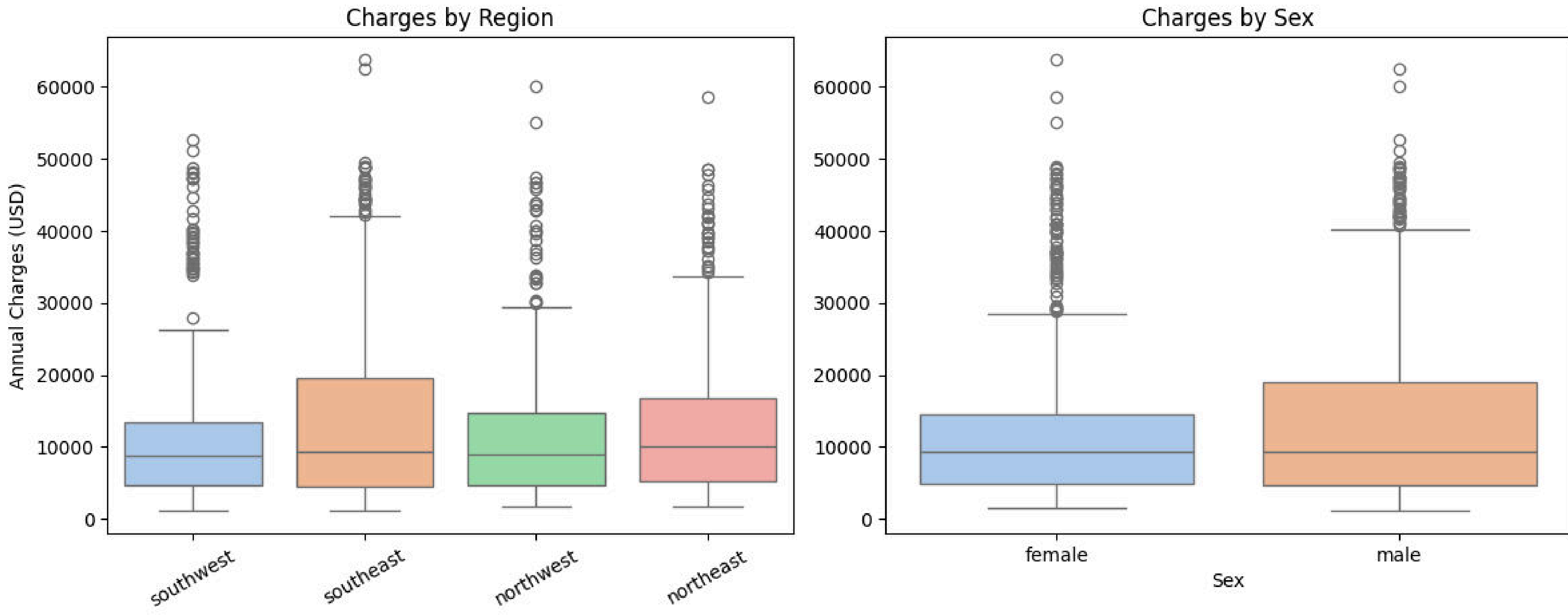


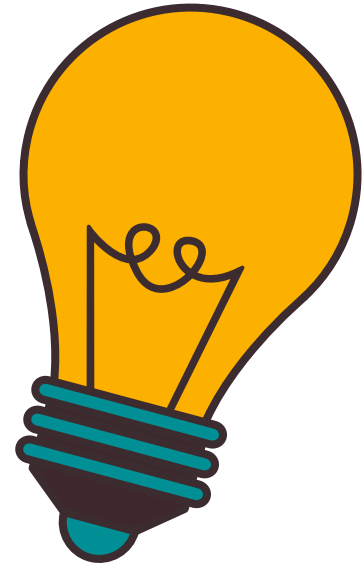
## BMI vs Charges



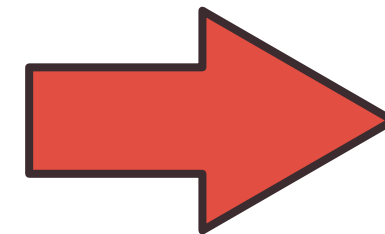
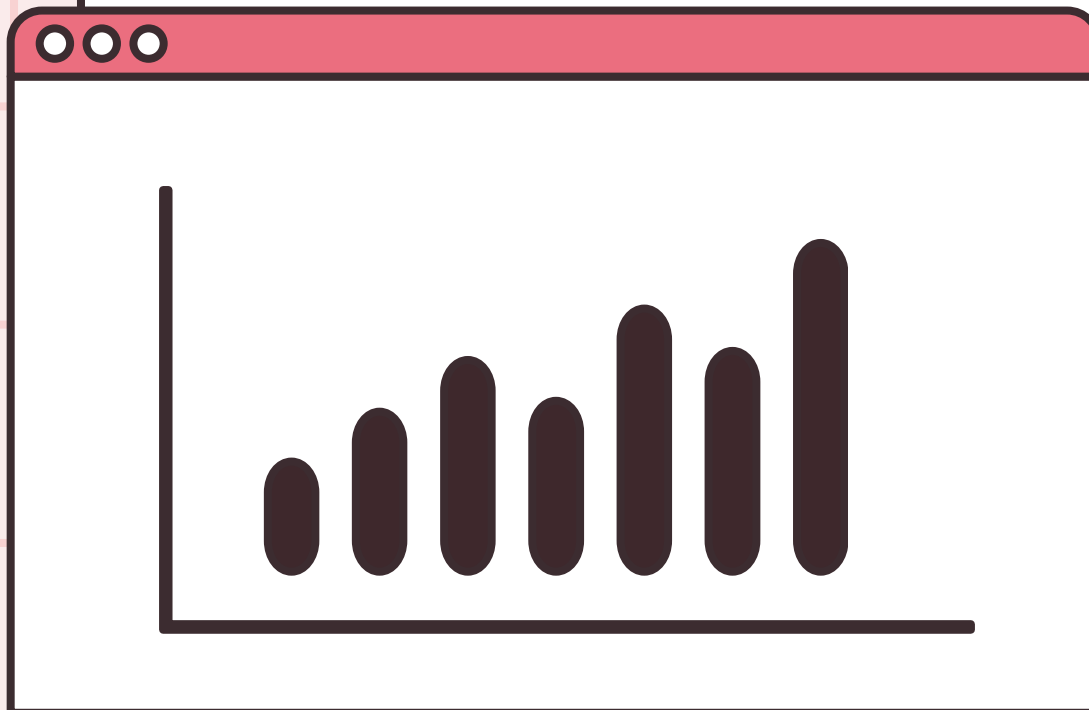


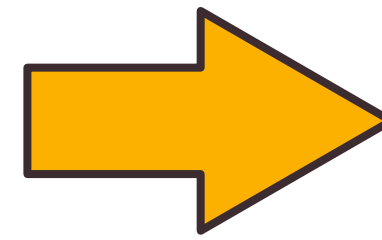
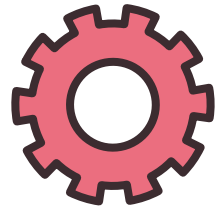
Region & sex show almost no difference in charges.





**Conclusion:**  
**Smoking** and **age** are the  
**strongest drivers of cost.**





# Preprocessing & Modeling

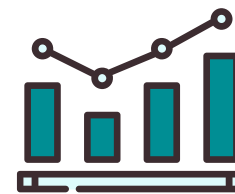


# Preprocessing



## Scales

Removed duplicates  
(1 row)



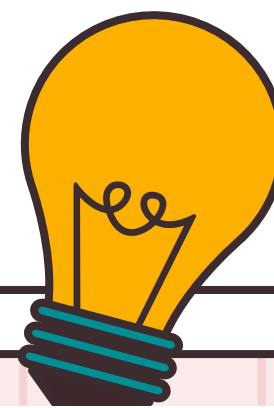
## Labels

One-hot encoded  
(sex, smoker, region)



## Data

Standardized numeric  
features  
(age, bmi, children)

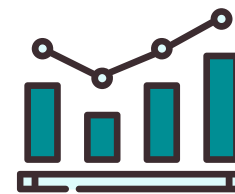


# Models trained



**OLS**

OLS (Linear Regression)



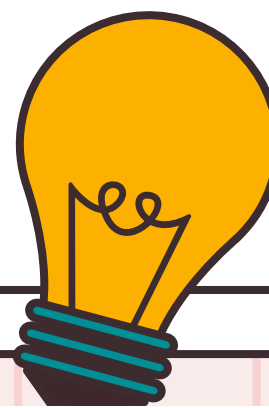
**Ridge**

Ridge ( $\alpha=1.0$ )

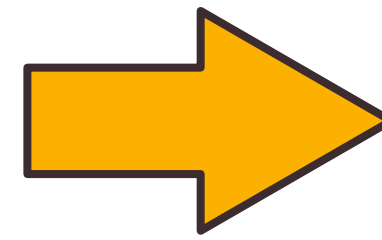
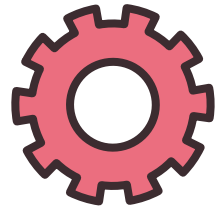


**lasso**

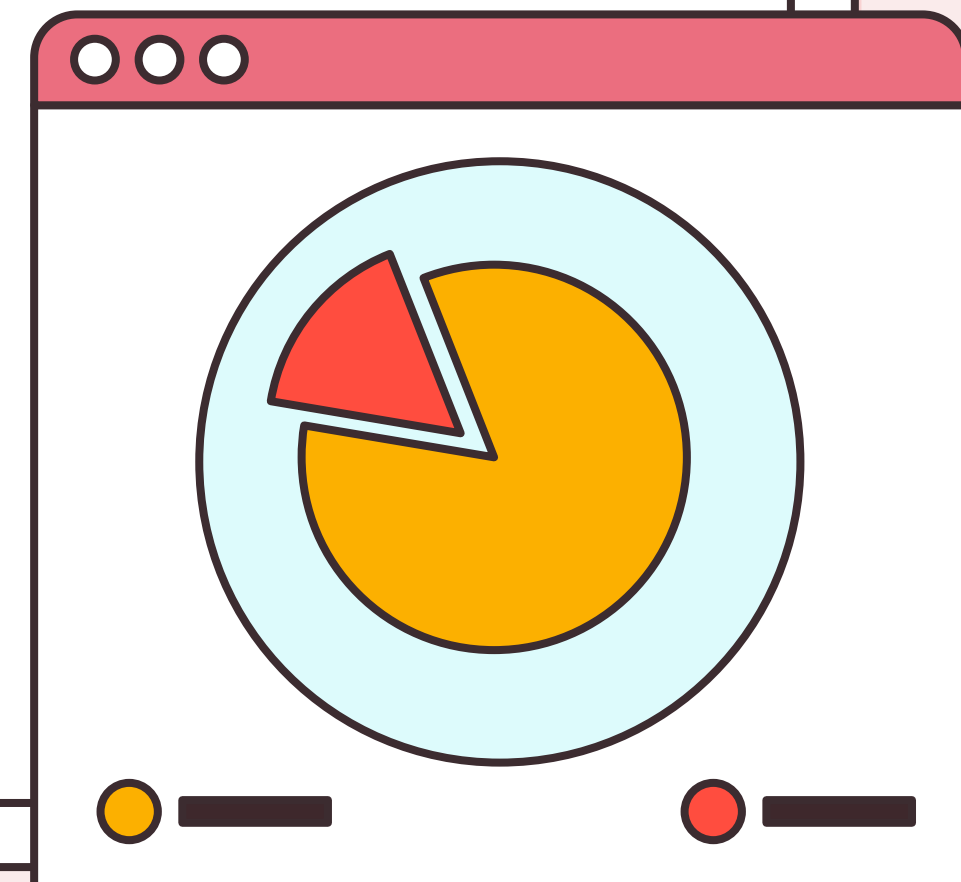
Lasso ( $\alpha=0.01$ )







# Model Performance & Interpretation



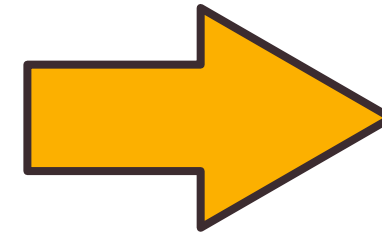
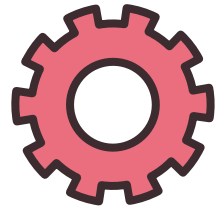
# Model Performance & Interpretation



Most important coefficients (OLS):

- `smoker_yes` → +23,000 USD
- `age` → +3,200 USD
- `bmi` → +2,900 USD (per 1 std dev)
- `sex_male, region` → very small

**Conclusion:** Smoking alone increases cost by \$23k, Age and BMI also matter, Region and sex are negligible.

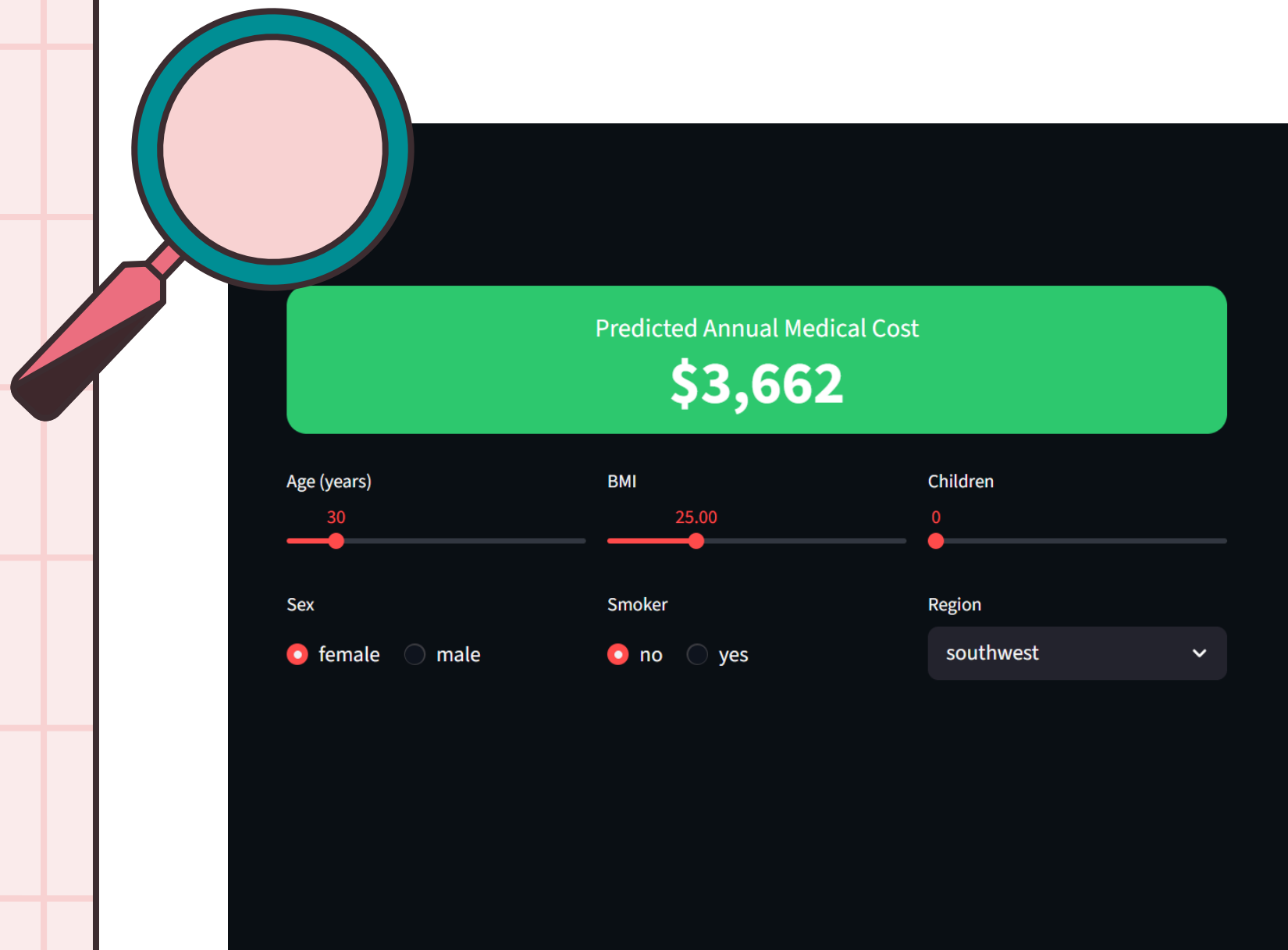


# Streamlit Dashboard





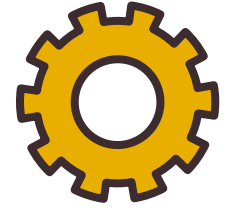
# Deployment - Streamlit Dashboard



What it does:

- Interactive web app (local or cloud)
- User inputs age, BMI, children, smoker, sex, region
- App returns predicted annual charge in real time.

Tech stack: Python, Streamlit, joblib, scikit-learn.



# Conclusion

- **Smoking** is the strongest predictor of medical costs, adding about \$23,000 per year.
- **Age** and **BMI** also increase costs, while **region** and **sex** have almost no effect.
- A simple linear regression model explains 78% of the variance, making it useful for insurance premium estimation.

